512	256	128	64	32	16	8	4	2	1

Rate conversions – x8 for bytes

bps	Kbps	Mbps	Gbps	Tbps	Pbps	Ebps
$\times 10^n$	3	6	9	12	15	18

Numerical Types

Unsigned: 0 → $2^n - 1$, Signed: $-(2^{n-1} - 1)$ → $2^{n-1} - 1$
Twos Complement: -2^{n-1} → $2^{n-1} - 1$
Binary → TC: Flip all bits and add 1

Hexadecimal: A: 10 | B: 11 | C: 12 | D: 13 | E: 14 | F: 15

Bit-fiddling

x << n (left): discards from left, pads right
x >> n (right): discards from right, pads left

Data rates

Data rate: raw physical layer speed
Throughput: measured successful rate, considers headers, idle times, retransmission
Goodput: rate of transmitting payload data, payload bits delivered ÷ transmission time
Video: dimensions × colour depth × frame rate
Colour count to depth d: num colours = 2^d
P/D/R: D is depth per RGB, ×3 for total depth

Delay

Transmission: time to push all bits onto the link – bits/bps
Propagation: time for a bit to travel across the link – distance/propagation speed
Queueing: time a packet waits in buffer before transmission

Packet Loss

Binomial Eg: P(k of n packets lost)	$\frac{n!}{k!(n-k)!}\, p^k(1-p)^{n-k}$
Binomial Complement Eg. P(at least 1 lost out of n)	$1-(1-p)^n$
Geometric Eg. P(k attempts until success)	$(1-p)^{k-1}p$ (k-1)→k not counting 1st success
Geometric Mean Eg. Expected attempts	$1/p$ (1-p)/p not counting 1st success

Circuit vs Packet Switching

Both experience propagation, transmission and processing delay.

Circuit Switching: Virtual private cable, routing decision made once at setup. Resources (buffers, processing capacity, bandwidth) are guaranteed. Additional call setup and teardown delay.

Pros	Cons
- Reserved resources shield the connection from other traffic. - Good for constant bit rate	- Reserved rate not utilised in pauses - Reserved rate may not suit during traffic peaks

Packet Switching: Breaks data into packets, no connection set up first, packets can be sent immediately. Additional queueing delay. Congestion is where packets are dropped from a full buffer

Pros	Cons
- Resources can be shared - Packets can be sent immediately - Good for variable bit rate	- Packets must include routing info - Packets might not arrive in order - No guarantee for delivery

UDP and TCP

User Datagram Protocol and Transmission Control Protocol are transport layer networking protocols. They both run on top of IPv4, generating packets to be payload for IPv4 packets. Both add port numbers. UDP receives individual datagrams, whereas TCP receives a byte stream. TCP requires a setup, lost data is retransmitted, bytes arrive in order, data can flow both ways. TCP sockets can share the same local port, since each connection is identified by (local IP, local port, remote IP, remote port).

Sockets

A socket is a software endpoint that combines an IP address, port number and transport protocol. They are bound to one port. Throws socket.gaierror for address errors, otherwise OSError
Socket Calls: A blocking (like accept() or select()) pauses the calling process until a relevant input arrives.
Blocking: accept() (can be changed), select(), recv[from]()
Non-blocking: bind(), listen(), send[to]() (only if there's enough buffer space)
listen(backlog=...): length of unprocessed incoming connection queue
accept(): returns a new socket to handle this connection and the remote address

Endianness

Little-endian: less significant bytes sit at lower addresses
Big-endian: the opposite

OSI Seven Layer Model

Single-hop scope: only exchange protocol data units (PDUs) between physically connected hosts. Applies to 1, 2.
End-to-end scope (E2E): logically talk to peer layer on other host even though data passes through other routers. Applies to 4-7.
Host implements 1-7, router implements 1-3
1) Physical: Physical data transmission (bits)
2) Link: Transfer of frames over one physical link
3) Network: Addressing and routing, end-to-end delivery of packets. Uses hop-by-hop through routers
4) Transport: Reliable, in-sequence end-to-end transfer of segments, error/flow/congestion control
5) Session: Manage communication sessions
6) Presentation: Translates between data representations on end hosts
7) Application: App support functions

TCP/IP Reference Model

(1) Physical layer (same as OSI), (2) Network Interface, (3) Internet, (4) Transport Layer, (5) Application
The Internet follows end-to-end principal: layers ≤3 (routers) kept simple, as they're main bottlenecks for speed, must process many packets fast, with limited hardware processing.
(2) Similar to link layer, exchanges datagrams between IP and physical link. Includes medium access control (MAC), framing, address resolution, might include link-layer error control and flow control
(3) Uses IP, PDUs are datagrams, specifies an addressing scheme, doesn't specify routing, no mechanisms for error, flow, congestion control
(4) Provides E2E to applications through socket interface. Where TCP and UDP sit.
(5) Accesses transport layer through socket interface. Application layer protocol examples: SMTP, HTTP, FTP, RTP

Service Providers and Users

Each protocol at a layer N is a service provider (SP) with an N-service, with service users (SU) being N+1 protocols. N-SDU (service data unit) is the data received from the above layer, PDU is what this layer constructs.
Confirmed service: SU at A requests a service S (s.request) → S's SP generates and sends PDUs to host B → B's SP informs B's SU is of the request (s.indication) → B's SU gives response to its SP (s.response) → A's SU is informed on B's response via A's SP (s.confirm)
Unconfirmed Service: Same until B's SU receives the s.indication, and no response happens → A's SU doesn't know if the request reached B
Confirmed Delivery Service: Same except B's SP generates the response, so A's SU has no info about the behaviour of B's SU

Multiplexing – combining independent data streams for several N SAPS so they can share an N-1 SAP and be restored on the other end. Service Access Points (SAP) are sockets/ports used for this. N protocol entity must make scheduling decisions, include SAP identifiers.

Fragmentation and Reassembly: N-layers splitting data into numbered fragments. Disadvantage is higher overhead.
Blocking: Transmitter waiting for multiple N-SDUs and putting them in one N-PDU with markers to save overhead

Sequence Numbers – An N-entity writes a sequence number into each new N-PDU header and then increments. This allows the receiver to detect duplicates, lost PDUs, and correctly reorder received PDUs.

Communication Systems

Source: Source signal
Format: formats source data into digital data
Source encoding (compression): encode data to reduce size.
Lossless: Encoded to be perfectly restored
Lossy: Compresses better, humans only perceive so much and wouldn't notice a difference. Often applied to video, audio, etc
Channel encoding: adds redundant bits to reduce transmission errors, reduce bit error probability. With single transmitter and receiver channels, separating source and channel encoding is optimal, not necessarily elsewhere.
Modulator: Maps bits to physical signals over a certain duration (symbol time)
Channel*: Physically transmits the signal
Demodulator*: Guesses original data
Channel decode: Uses redundancy from channel encoding to correct errors
Source decode: Uncompresses data
Format: Converts data to output
*Receives analogue waveforms

Baseband Transmission

Medium cannot be shared
NZR Encoding: Each bit value is assigned a voltage level and transmitted for symbol time seconds. Inverse is symbol rate, equal to bit rate.
→ Bipolar: bits mapped to a +ve and -ve voltage
→ Unipolar: bits mapped to a 0 and +ve voltage
M-ary Signalling: Groups of k bits are mapped to $M = 2^k$ different voltage levels and transmitted over symbol time. Symbol rate ≠ bit rate. These levels will be closer together due to transmit power restrictions, increasing bit error rate.

Attenuation – loss of signal power

Ratio of output signal power (P_{tx}) to input signal power (P_{rx}). $P_{rx} = P_{tx}/\eta$
Since η can be very large, its often represented in dB, $\eta dB = 10 \times \log_{10}(\eta)$. Inverse = 10**(ηdB/10)
Attenuation can vary for different frequencies. A signal can be decomposed into sine and cosine waves (Fourier analysis) allowing the channel to treat frequencies differently.
When attenuation is frequency-independent, r(t) = s(t)/ η where r(t), s(t) are receiver and sender signals

Thermal Noise

Additive white Gaussian noise

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x}{\sigma}\right)^2\right]$$

With thermal noise,
reciever will observe r(t) + f(x)

Passband transmission

Also known as frequency-division multiplexing.
Data is transmitted by modulating a sinusoidal carrier of a certain centre frequency (CF). Resulting signal occupies a certain bandwidth around the CF.

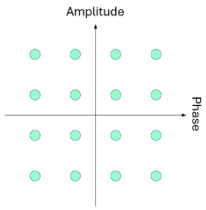
$$s(t) = A(t) \cdot \cos\left[(f_c + f(t)) \cdot t + \phi(t)\right], \quad t \in [0, T]$$

T = symbol duration, A(t) = amplitude, f(t) = frequency offset, ϕ(t) = phase, f_c = centre frequency. Three "pure" forms: amplitude, frequency, phase modulation/shift keying, where only one element is varied.

ASK	$s_i(t) = A_i \cos(f_c \cdot t) \quad (0 \le t \le T)$
FSK	$s_i(t) = A\cos((f_c + f_i) \cdot t) \quad (0 \le t \le T)$
PSK	$s_i(t) = A\cos(f_c \cdot t + \phi_i) \quad (0 \le t \le T)$

Quadrature Amplitude Modulation

Combination of ASK and PSK, usually applied with 16, 64 or 256 constellation points, groups of 4, 6 or 8 bits are mapped to one QAM symbol

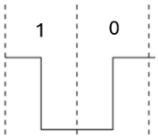


Synchronization

Sender and receiver (S+R) need to agree on: unit time length, symbol period. Usually no access to common time or frequency reference, both have their own local references provided through crystal oscillators. Oscillator frequency depends on time (oscillator aging) and environment (temp, pressure, supply voltage), therefore is different between S+R. R must learn reference of S – needs to acquire synchronization initially and track/maintain it during packet reception. In passband transmission, S+R agree on centre frequency. Frequency mismatch potential causes: oscillator mismatch, doppler shift. With PSK and QAM signal phase also needs tracking. Carrier synchronization is combination of frequency and phase syncing.
Symbol/bit Synchronization: S transmits at its own pace using local clock to generate symbol duration. R extracts

symbol duration info from signal. Long streams require continuous synchronization, as long durations of same input can cause loss of sync.

Manchester Encoding



Receiver internal clock samples incoming signal with a much higher freq. Eg: freq = 16, for a '0' bit the sample would be 00000000 11111111, uneven amount of 0s and 1s, so clock is readjusted. Ethernet's 64 bit long preamble is for this.

Frame Synchronization

In PHY layer, sync field for bit-syncing, start-frame-delimiter (SFD) to indicate where data starts. Some methods: time gap (medium is idle between frames), code violations (deliberate violations eg symbol durations without signal change in Manchester), length field (dedicated field to indicate length of frame, additional methods needed to deal with bit errors).

Start/end flags: Flags (specific bit sequence) can be used to mark the start and end of a frame. Example: HDLC uses 01111110. Payload data may contain this, sender must alter any flag-like data. Bit-stuffing is where the sender inserts a '0' bit after 5 consecutive '1's. If a '0' is detected, it is removed, otherwise this is a flag. Transmission errors can create an accidental flag, so additional mechanisms like time gaps are needed.

Local Area Networks (LANs)

Packet-switched networks, packets = frames. They offer: shared transmission medium to multiple stations, low cost for station attachment, higher rates than usual in wide-area networks. Typically specify physical layer and medium access control, optional link layer providing error control (detection, correction and retransmission) and flow control (stopping fast transmitters from overwhelming slow receivers). Stations are attached via tap line to bus. Bus is a broadcast medium, signals heard by all stations. Parallel transmissions collide garbling frames. Frames include destination address. Bus length limited to hundreds of metres to prevent excessive signal attenuation. Star topology is where stations are attached to a central unit which can act as a repeater (relays incoming signals to all other outputs) or bridge/switch (only relays to destination).

Medium Access Control

MAC protocols allow many users to coordinate access to a common channel. Often regarded as sub-layer between PHY and link-layer. We're given: number of users/stations and a shared channel.

Design Goals: Small medium access delay (time between frame arrival and start of transmission), fairness and re-use of unused resources, efficiency (low overhead, high throughput), stability (increased load ≠ decreased throughput).

Orthogonal Schemes

Behaviour of one station doesn't influence other stations, allowing for collision free transmission. DMA → Division Multiple Access

FDMA (Frequency): bandwidth split into sub-channels with guard-bands, then assigned to stations. No medium access delay, transmission can instantly start with no collision risk.
→ Advantages: Parallel transmission, no time synchronization needed between transmitters
→ Disadvantages: Needs tuneable or N many receivers, frequency synchronization needed, unused channels inaccessible, coordination and shared state needed for allocating subchannels.
→ Good for CBR, bad for VBR

TDMA (Time): Each station uses whole frequency band (except guard bands) at certain times. Time split into superframes of duration T_{SF} which are split into N time-slots with guard times. One or more time-slots are assigned to stations. Stations must be time-synced to avoid overlap. Medium access delay = $T_{SF}/2$. Transmission delay = 1/N. E[Completion Time] = E[Access Delay] + E[Transm. Delay] ≤ 1s, for N > 2. With TDMA we can start later and finish sooner than with FDMA. Advantages: easier to give different stations different amounts of the channel (giving more time slots), better completion times. Disadvantages: Tight time-sync needed between stations, high access delay, can't reuse unused time slots, coordination needed.
→ Good for CBR, bad for VBR

Random Access Protocols

Don't: attempt to reserve channel resources for longer time, require central station, access medium at predictable times. Often low complexity, typically involve random element. Accept risk of collisions to save coordination overhead, depends on traffic load.

Aloha: Assumptions – N uncoordinated transmitters, one receiver. On frame arrival, checksum is appended then immediate frame transmission, an acknowledgement timer is started. When ack expires, transmitter enters backoff mode: chooses random backoff time (to reduce collision chance) → waits → timer expires → frame retransmission. Frame is dropped after exceeding number of failed trials. Frames stored in queue, next frame is transmitted after random backoff. Frame vulnerability period is two frames long.

Carrier Sense Multiple Access (CSMA)

Assumption: All stations can determine the state of the medium near instantly (carrier sensing/clear channel assessment). Approach: listen-before-talk, stations check if channel is busy, backs off if so. Reduces collision rate/impact. Useful when prop time is small compared to frame length, fast to notice transmission. Collisions only when two transmissions with time difference < prop delay.

Nonpersistent: CS → random backoff time

P-persistent: Time slots chosen to accommodate time for prop delay and carrier sensing → CS → when idle, probability p = transmit now, probability p-1, wait one slot and repeat.

Choosing p: Close to 1 → behaves like 1-persistent, fast to grab idle channel but more collision prone with several waiting stations. Close to 0 → behaves like nonpersistent, collisions less likely, but channel may be idle for longer.

1-persistent (CD): Chosen for Ethernet. CS → stations send immediately when medium is idle → likely collisions, sender monitors and if detected → transmission aborted and jamming signal sent to all stations → collision resolution

Ethernet: Address fields 48 bits (so transmitter can always receive a jamming signal). min, max payload = 46, 1500 bytes. Addresses: Broadcast = FF:..., Multicast = odd leftmost byte (ends in 1), Unicast = even leftmost byte (ends in 0). Length/type, ≥ 0x0600 → type, else length.

Half-duplex: Broadcast Topology, stations share medium capacity, MAC needed. In Ethernet, Bus topology and Hub-based topology. Only used in 10Mbps PHYs. Uses CSMA/CD. MAC Protocol counts collisions, drops when >16. Backoff time: random number from backoff window [0, 2**min{10,coll} -1], × slot time. Short access delay under light load, high collision rate with heavy load, should not be operated beyond 50-60%.

Full-duplex: Switched Topologies. Stations attached to switches via point-to-point links, no MAC needed. Links have separate cables allowing parallel transmission and receiving. Used for 100+ Mbps. "Switched Ethernet".

Coupling LANS: physically connecting two separate LANs. Requires coupling devices (PHY layer: repeaters, hubs. MAC layer: bridges, layer-2 switches. Network layer: router. Application layer: gateway)

Repeaters/Hubs

Hubs: Centralised repeaters, broadcast incoming signals to all other ports. Stations have a transmit and receive line. Only one station can transmit at a time without collisions. Advantage over bus: can work with a cut cable.

Repeaters: amplify signal at analogue level, adding noise.

Regenerating Repeaters: reads signal symbol-by-symbol and transmits new, clean waveform.

Bridges/Switches

Bridges: interconnect (several) LANs on MAC layer. Process MAC related fields, repeaters/hubs don't. Bridge knows which stations in which LANs from frame source addresses. When receiving a frame with unobserved dst, it retransmits everywhere apart from src. Dangerous if loops are present. Benefits: reliability – LAN failures don't affect others, performance – local LAN interactions don't affect other LANs, security: local LAN traffic can't be eavesdropped.

Switches: Centralised element, only forwards frame to correct output. Stations attached full-duplex. Can process frames in parallel, increased network capacity. Frames arriving in parallel for same dst are buffered. Can accept up to (num stations) parallel transmissions.

IPV4: Datagram delivery - Connectionless, Unacknowledged (no receive confirmation), Unreliable (on IP level no retransmissions), Unordered. Best effort.

Classless Inter-Domain Routing

Network specified by 32 bit network address and netmask, specifying which bits belong to network-id and host-id.

11111111.11111111.11111110.00000000	/23

Host address: 0 = whole network, 255 (8 '1' bits) = broadcast address

Packet Processing: Check IP version field → check IP header checksum → protocol demultiplexing if packet destined to this device → deliver payload to software entity implementing the protocol in the IP header END→ (drop if packet forwarding not enabled) otherwise check if destination is directly reachable (if so, deliver then END) → consult forwarding table and decrement TTL → recompute header checksum → deliver.

Forwarding table is maintained by a routing daemon. Each entry contains: Destination IP address, Information about next hop either: IP address of next-hop router (must be directly reachable) or IP address of directly-connected network (network address/netmask), Specification of outgoing interface. Forwarding table lookup for a packet with destination IP address dst: 1) Look for entry that is a network address matching dst – if found, send packet to given next hop / outgoing interface. 2) Look for special default entry – if found, send packet to indicated next hop (the default router). Otherwise drop packet, possibly send ICMP message back to original sender of datagram. dst matches a table entry if (dst AND < /k – netmask >) == (a.b.c.d AND < /k – netmask >), leftmost k bits of the destination address and network address match.
Some routers in the core: Do not have a default router, Are the default routers of other routers, Must know (almost) all the Internet networks. Fragmentation: destination receives first fragment → allocates 65,550 bytes and starts timer → receives fragments until one has MF=0 (more fragments) → combine last packet's fragment offset with its own length to find end of packet.

Address Resolution Protocol – ARP

I an ethernet, stations have their own 48-bit MAC addresses. IP datagrams are encapsulated into Ethernet frames. Ethernet station picks up a frame only if the destination MAC address matches.
ARP determines MAC address for given IP address, it is dynamic. Advantage: nodes can be be moved.
Disadvantage: a separate protocol is needed, additional complexity and bandwidth.
Each station in the ethernet maintains an ARP cache, which stores the known mappings from IP to MAC addresses. If station doesn't know a MAC address: broadcast ARP-request with A's IP and MAC and B's IP → B receives request, stores A's IP and MAC in its ARP cache → responds with unicast ARP-reply including A and B's IP and MAC → A stores B's IP and MAC. Entries are typically removed 20 mins after creation (easier than validating cache entries, eg a crashing station has no way to communicate).

Internet Control Message Protocol – ICMP

Allows routes to inform sender about unusual situations (no route, unreachable destination, fragmentation required but DF is set). They are encapsulated into IP datagrams, no additional mechanisms, IP sending host must not rely on ICMP messages

type	code	Meaning
0	0	Echo reply
3	0	Destination network unreachable
3	1	Destination host unreachable
3	2	Destination protocol unreachable
3	3	Destination port unreachable
3	4	Fragmentation required, but DF bit set
3	6	Destination network unknown
3	7	Destination host unknown
4	0	Source quench (Congestion control)
8	0	Echo request
11	0	TTL expired in transit
11	1	Fragment reassembly time exceeded